Simulation-Based Electromagnetic Analysis of TEM, TE and TM Propagation Modes in Microwave Transmission Structures

➢ CO Mapping: CO1

Bloom's Level: K4 – Analyze

WK Level: WK3 – Fundamental propagation modes

Aim:

To develop electromagnetic simulation models for TEM, TE and TM propagation modes in suitable microwave transmission structures.

Software Required:

MATLAB / CST Studio Suite / Ansys HFSS / COMSOL

Matlab Code:

```
clc; clear; close all;
c=3e8; f=10e9;
a=22.86e-3; b=10.16e-3;
fc=c/(2*a);
ft=c/2*sqrt((1/a)^2+(1/b)^2);
k=2*pi*f/c;
beta=sqrt(k^2-(2*pi*fc/c)^2);
al=sqrt((2*pi*ft/c)^2-k^2);
fprintf('Suraj | Reg: 192412442\n');
fprintf('TE10 fc=%.3f GHz\n',fc/1e9);
fprintf('Beta=%.2f rad/m\n',beta);
x=linspace(0,a,40); y=linspace(0,b,40);
[X,Y]=meshgrid(x,y);
figure('Color','w');
sgtitle({'Suraj | Reg No: 192412442', ...
    'Rectangular Waveguide Analysis'});
subplot(2,2,1); imagesc(sin(pi*X/a)); colorbar; title('TE10 E-Field');
subplot(2,2,2); imagesc(cos(pi*X/a)); colorbar; title('TE10 H-Field');
subplot(2,2,3); imagesc(sin(pi*X/a).*sin(pi*Y/b)); colorbar; title('TM11 E-Field');
subplot(2,2,4); imagesc(sin(pi*X/a)); colorbar; title('TM11 H-Field Pattern');
```

colormap(jet);

F=linspace(1e9,20e9,80); B=nan(size(F)); q=F>fc;

B(q)=2*pi*F(q)/c.*sqrt(1-(fc./F(q)).^2);

figure('Color','w'); plot(F/1e9,B,'b','LineWidth',2); grid on;

xlabel('Frequency (GHz)'); ylabel('Beta (rad/m)');

title('TE10 Propagation Constant');

Formula to be used:

• Waveguide cutoff frequency: fc(mn) = (c/2)√[(m/a)2 + (n/b)2].

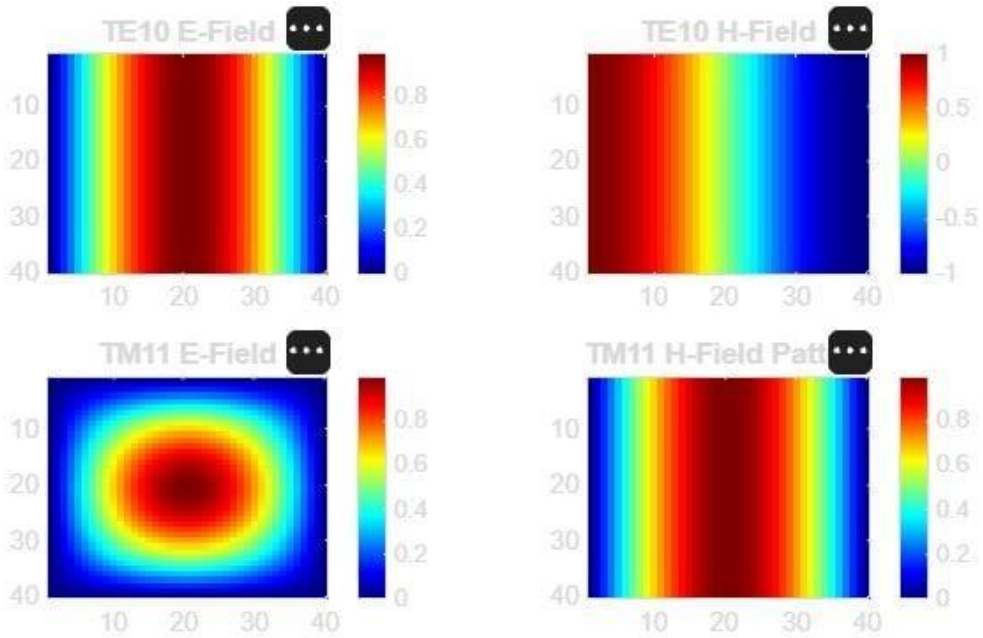
• For TE10 mode in a rectangular waveguide: fc10 = c/(2a).

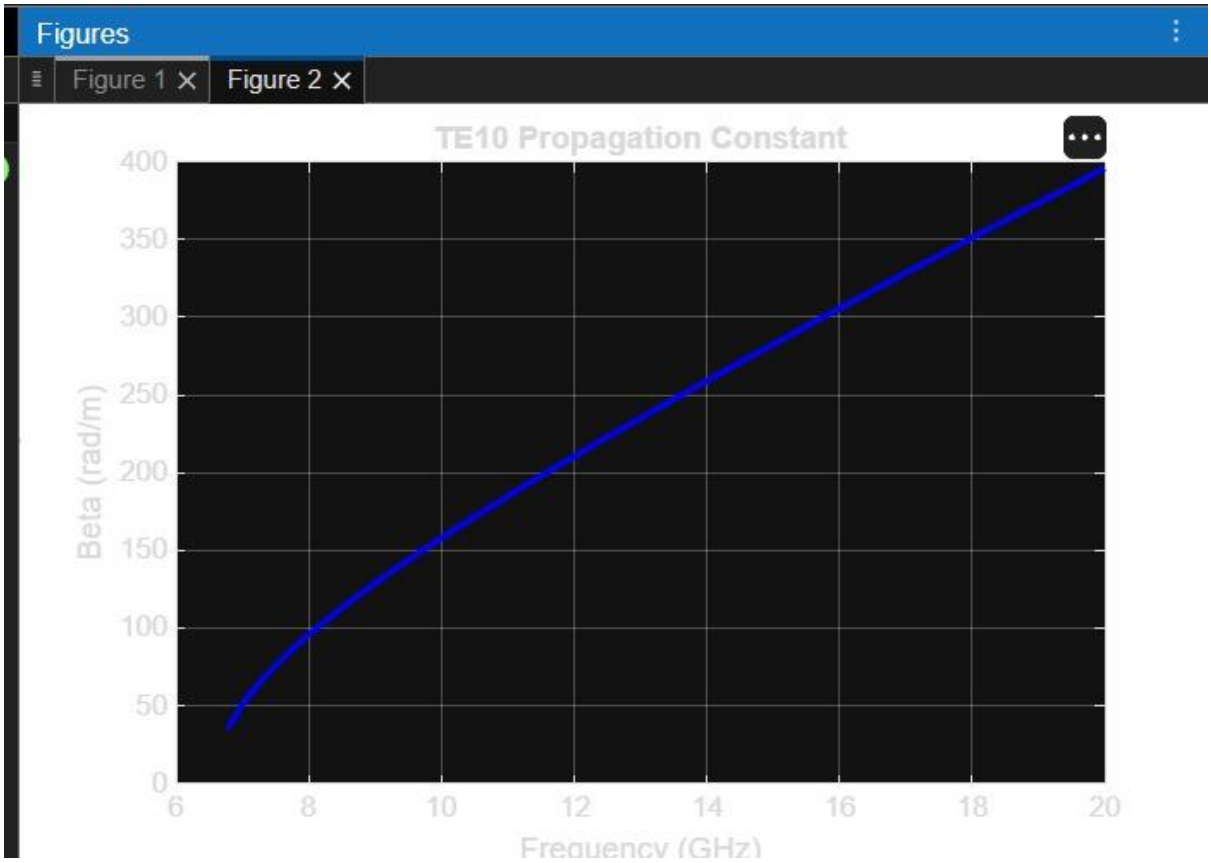
• Above cutoff, propagation constant: β = k0√[1 − (fc/f)2], where k0 = 2πf/c.

• Percentage deviation = |Practical − Theoretical| / |Theoretical| × 100.

• TEM mode has no cutoff frequency in an ideal TEM transmission line; TE/TM modes have cutoff.

Output:





Result:

Thus the Simulation-Based Electromagnetic Analysis of TEM, TE and TM Propagation Modes in Microwave Transmission Structures